

Natural History of Trek 12-15 at Philmont Scout Ranch

Executive summary

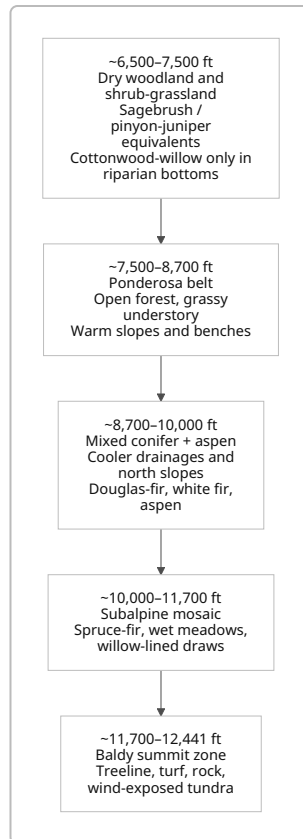
Trek 12-15 crosses one of the sharpest ecological gradients available on a Philmont itinerary. Over the course of the trek, hikers move from lower, warmer mesas and canyon bottoms around Ponil and Indian Writings through ponderosa pine and mixed-conifer belts, then into the high subalpine and near-alpine environments around Baldy Town and Baldy Mountain. That elevational range mirrors the broader Carson National Forest transition from sagebrush and lower montane woodland to alpine and tundra systems above treeline. In practical field terms, the single most important organizing idea for this guide is that **elevation, slope aspect, and substrate control moisture**, and moisture then controls what grows where. South-facing, rocky, fast-draining slopes stay open and warm longer; north-facing slopes, shaded drainages, and deeper colluvial or alluvial soils support denser forest, taller shrubs, and longer-lasting late-season bloom. ¹

For Trek 12-15 specifically, expect five major ecological settings: lower-elevation dry woodland/shrub-grassland near Ponil and Indian Writings; open ponderosa pine forest through much of the mid-elevation travel; cooler mixed-conifer forest around Harlan, Santa Claus, and the approach to Baldy; subalpine spruce-fir, aspen, and meadow mosaics around Baldy Town and French Henry; and a short but important above-treeline zone on Baldy Mountain where wind, frost, and shallow soils dominate. Wildlife shifts with those zones: elk, mule deer, black bear, and wild turkey are most broadly distributed; Abert's squirrels and Steller's jays key into conifer forest; broad-tailed hummingbirds track flowering meadows and forest edges; pikas, marmots, and Clark's nutcrackers become much more plausible near the highest ground. ²

The itinerary you shared places the trek through Harlan, Devil's Wash Basin, Cimarron River, Santa Claus, Baldy Town, Pueblano, Elkhorn, Ponil, and Indian Writings before returning to base. That sequence is what the day-by-day segment analysis below is keyed to.

Ecological framework

Philmont sits on the east side of the Cimarron Range within the Sangre de Cristo Mountains, and the neighboring Carson National Forest summarizes the regional setting well: elevations run from about 6,500 feet to over 13,000 feet, with ecosystems ranging from sagebrush at low elevations to alpine and tundra at the highest elevations. Nearby southern Rockies wilderness descriptions add the practical field pattern advanced hikers actually use: lower elevations support pinyon-juniper and ponderosa; middle elevations grade into mixed conifer and aspen; higher elevations support spruce-fir; and alpine tundra occurs above treeline, which in this region is roughly around 12,000 feet. ³



That diagram is a field-use simplification rather than a hard ecological boundary map; local moisture can shift the bands hundreds of feet up or down. For example, the Carson region’s “Colorado Borderlands” pairs rolling grasslands with surrounding conifer, ponderosa, and aspen stands, while Valle Vidal pages emphasize spruce, aspen, and fir in cooler sites and ponderosa in other sectors. That is exactly the kind of patch mosaic hikers should expect in north-country Philmont: the same nominal elevation can carry different vegetation depending on aspect, drainage, burn history, and soil depth. 4

Geology matters because rock type and geomorphology set the soil template. Coarse, rocky, rapidly drained upland slopes tend to favor open forest or shrub-grassland; deeper slopewash, meadow alluvium, or riparian fill stores more water and supports denser herb cover, willow, cottonwood, and lingering bloom. Around the highest parts of the trek, shallow, wind-stripped soils and frost exposure suppress trees and favor low turf, cushiony herbs, sedges, and very short growing seasons. That relationship between elevation, exposure, and ecological change is a standard southern Rockies pattern and is especially visible on a route like Trek 12-15 because it climbs so abruptly into Baldy country and then drops again toward Ponil. 5

Fire is now part of the ecological framework, not a side story. The Ponil Complex burned a very large area of Philmont’s north country and adjacent Valle Vidal in 2002, and subsequent flooding and erosion damaged riparian zones, altered channel form, increased sediment transport, raised stream temperature where shade was lost, and harmed aquatic habitat. On the ground, hikers should expect burn mosaics rather than a simple “burned/unburned” binary: dense young aspen or shrub fields in places, open snag-rich hillsides in others, and recovering grass/forb communities where forest canopy was removed. 6

Vegetation communities and field flora

The most important trees on this trek, in order of increasing elevation, are likely to be pinyon-juniper associates and shrubs in the lower dry country, then ponderosa pine, then mixed-conifer species such as Douglas-fir and white fir, then quaking aspen and Engelmann spruce in the coolest higher settings, with true summit terrain on Baldy supporting only scattered krummholz-like stunting or treeline remnants before giving way to alpine turf and rock. The adjacent Carson pages are explicit that nearby regional settings include ponderosa, aspen, fir, spruce, and alpine/tundra communities; the Philmont north-country analog is very close. ⁷

For field recognition, ponderosa pine is the easiest anchor species: look for tall, open-grown trunks with thick orange to cinnamon plates on mature trees, broad spacing, and a sunnier, grassy understory. Mixed conifer is denser and cooler; the forest feels darker, with more layered canopy, more down wood, and more needle litter. Aspen stands show smooth white-green bark, black scars, broad flat leaves, and a conspicuously brighter understory where light reaches the ground. Spruce-fir forest feels immediately colder and tighter: spires, shade, moist duff, and meadow edges with sedges or wet-loving forbs. Those “forest feel” cues are often more useful on trail than trying to key every conifer from needles alone. ⁸

Shrub structure also tells you where you are. In warmer and more disturbed mid-elevation sites, expect more open shrub fields and locust/oak-type edges; in colder meadows and draws, expect willow-lined wet zones. New Mexico locust is a good “disturbance and edge” indicator in the southern Rockies and is notable because its showy pinkish flower clusters and spiny stems make it easy to remember once learned; it often becomes obvious along openings and recovering slopes. Riparian thickets, by contrast, read as willow/cottonwood habitat because the vegetation follows the water rather than the bedrock. ⁹

Wildflower diversity will be highest where light and moisture coincide: meadow margins, post-fire openings, stream terraces, aspen edges, and the fringes of mixed-conifer stands. The broad-tailed hummingbird's association with flower-rich montane and subalpine meadows is a useful indicator of where nectar plants are concentrated. In practical hiking terms, if you are in open, sunny subalpine meadow near timberline or around aspens/spruce edges, that is where to slow down for columbines, paintbrushes, lupines, asters, and gilia-type blooms, depending on recent moisture. ¹⁰

The seasonality is strongly elevation-dependent. The June-to-September pattern below is therefore best treated as an informed field inference for this route rather than a fixed calendar, because snowpack and monsoon timing can shift bloom by several weeks. Still, the pattern is reliable enough for guide use: lower elevations lead the season, subalpine meadows peak later, and Baldy's summit flora is always compressed into the shortest window. This interpretation follows the Carson southern-Rockies elevation framework and species-habitat summaries for subalpine meadow birds and alpine-adjacent communities. ¹¹

Community	Likely indicator plants	Approximate trek elevation	Best viewing window	What to notice in the field
Dry lower woodland / shrub-grassland	Sagebrushy shrubs, juniper-pinyon associates, grasses, riparian cottonwood/willow only where water persists	~6,500–7,500 ft	June–September	Wide spacing, bright light, rocky soils, hotter aspect
Ponderosa belt	Ponderosa pine, grassy understory, edge shrubs including locust/oak associates	~7,500–8,700 ft	June–September	Orange bark, open canopy, park-like spacing
Mixed conifer	Douglas-fir, white fir, ponderosa overlap, aspen patches	~8,700–10,000 ft	Late June–September	Cooler air, denser shade, more litter and logs
Subalpine mosaic	Aspen, Engelmann spruce, fir, sedges, meadow forbs, willow in wet draws	~10,000–11,700 ft	July–August	Meadow/forest patchwork, seeps, hummocky wet ground
Baldy summit zone	Low alpine herbs, sedges, turf, lichens, wind-pruned vegetation	~11,700–12,441 ft	July–early September	Short plants, exposed rock, strong wind, abrupt treeline

This table synthesizes the Carson National Forest elevation/ecosystem overview, adjacent southern Rockies wilderness vegetation descriptions, and the higher-elevation Baldy context relevant to Trek 12-15. ¹²

A short field key that works well on Philmont without a botanical manual is this. If the forest is **open, warm, and orange-barked**, think ponderosa. If it is **cool, denser, and patchy with aspens**, think mixed conifer. If it is **dark, moist, and meadow-fringed**, think spruce-fir/subalpine. If there are **no trees and plants hug the ground**, you are at or above treeline. For flowers, the easiest useful distinction is habitat first and flower second: riparian bloom, forest-edge bloom, burn-opening bloom, or high-meadow bloom. That approach is faster and more accurate on trail than trying to memorize every species in isolation. ¹³

Fauna and seasonal natural history

Large mammals likely to matter most to hikers are elk, mule deer, black bear, and, less commonly observed but ecologically present, mountain lion and smaller carnivores. The Valle Vidal and broader Carson materials emphasize the area's major elk herd and broad big-game assemblage, while adjacent high-country wilderness summaries add marmot, pika, and high-elevation birds to the upper-zone picture. The practical field takeaway is that **ungulate sign becomes more important than direct sightings**: beds, droppings, tracks, rubbed saplings, clipped meadow edges, and heavily used crossings are often more reliable than actually seeing the animals. ¹⁴

Elk behavior shapes management in the region strongly enough that Carson National Forest closes parts of Valle Vidal seasonally to protect wintering habitat and calving season. That tells hikers something biologically important even outside the closure area: early summer is a sensitive reproductive period, and meadows, edge habitat, and cover near water are not just scenic—they are functional habitat. When you move quietly at dawn or dusk through meadow margins near Baldy country or north-country drainages, you are walking through the highest-probability elk habitat on the trek. 15

Birding on Trek 12-15 is best organized by elevation. In conifer forest, expect mountain chickadees and jays as the “background” acoustic community; in high subalpine and near-treeline settings, Clark’s nutcrackers become especially meaningful because they are tied to high conifer systems and move between elevations seasonally; in flower-rich openings and meadows, broad-tailed hummingbirds are a strong possibility in summer. Steller’s jay and wild turkey are also plausible and often more detectable than rarer specialties. 16

The broad-tailed hummingbird is one of the best natural-history species for teaching habitat linkage on this route. Its breeding habitat includes subalpine meadows, montane valleys, and aspen/spruce settings, and its activity tracks nectar availability. In field-guide terms, if the group sees repeated hummingbird use of a meadow edge, that is usually a sign to stop and inspect the flowers rather than merely admire the bird. The bird is telling you where the bloom concentration is. 17

Near the highest elevations, mammals become smaller but ecologically sharper indicators. Pikas, marmots, and similar high-country species are tied to talus, rock piles, and cold alpine-adjacent settings in neighboring peak and wilderness environments very similar to Baldy. If hikers go above treeline and begin seeing persistent rock-talus habitat, that is where to look and listen for these species; if they stay below treeline in continuous forest, the probability drops sharply. 18

Aquatic and riparian natural history on this trek is regionally important but currently less predictable than the upland story because wildfire and flooding have altered stream habitat in north-country watersheds. Rio Grande cutthroat trout remain a flagship native species in the region, and the Valle Vidal explicitly highlights them, but post-fire effects on shade, sediment, flood dynamics, and stream temperature have made recovery uneven in burned watersheds. Hikers should treat streams as biologically sensitive corridors first and scenic lunch stops second. 19

I do not have a current, Philmont-specific publicly indexed source set that cleanly documents the exact reptile, amphibian, and insect assemblage for every Trek 12-15 segment. The high-confidence field statement is more modest: lower and warmer rocky country is the more likely zone for reptiles; cooler wet meadows and riparian edges are where amphibian activity is more likely after rain; and insect abundance rises sharply in wet meadows, willow edges, and monsoon periods, especially where bloom is concentrated. That is the right way to help hikers look, even where exact species lists remain incomplete in the sources I reviewed. 20

Trek 12-15 segment notes

The itinerary sequence below follows the Trek 12-15 route you shared and treats the camps and travel legs as ecological transitions rather than just mileage chunks. Because I do not have a publicly indexed, camp-by-camp vegetation survey for every Philmont segment, the species and communities listed are **most likely** rather than guaranteed on a given day. 21

Segment	Dominant ecological transition	Most likely things to observe	Why it matters
Base to Harlan	Lower montane into ponderosa/mixed-conifer transition	Open pine giving way to cooler forest pockets; rising moisture with elevation	Good first lesson in aspect and elevation control
Harlan to Devil's Wash Basin	Mid- to higher-elevation forest and basin country	Mixed conifer, shaded drainages, more meadow influence in openings	Shows how basins and draws retain moisture
Devil's Wash Basin to Cimarron River	Descent toward riparian corridor	Shrubs and grasses increase; riparian vegetation concentrates	Best place to explain why water overrides bedrock locally
Cimarron River to Santa Claus	Climb back into cooler montane forest	Ponderosa shifting to mixed conifer; edge bloom in openings	Good comparison with previous river-bottom vegetation
Santa Claus to Baldy Town	Upper montane to subalpine	Aspen, fir/spruce influence, meadows, cooler air	Transition into high-country ecology
Baldy Town layover / Baldy ascent	Subalpine to above treeline	Treeline, alpine turf, talus, pika/marmot probability rises	Best day for reading climate directly from vegetation
Baldy Town to Pueblano via French Henry	High subalpine down through disturbed/mining terrain	Meadow/forest patchwork, recovery vegetation, mine-disturbed substrate	Ideal geology-vegetation teaching day
Pueblano to Elkhorn	North-country forest and burn mosaic	Aspen regrowth, shrubfields, snag/opening contrasts in places	Fire history becomes legible
Elkhorn to Ponil	Drop toward lower montane / dry-country edge	Forest opens, grasses increase, warmer and drier feel	Shows the reverse side of the elevation gradient
Ponil to Indian Writings	Lower-elevation dry woodland and cultural corridor	Open woodland, shrub-grassland, riparian patches, rock surfaces	Best day to link ecology and archaeology
Indian Writings to Base	Similar lower-montane return	Dry-country species and human-disturbed communities	Useful recap of the entire gradient

This summary table is synthesized from your itinerary, the Carson National Forest ecosystem framework, and the adjacent-regional vegetation descriptions that most closely match high-country Philmont. ²²

Base to Harlan

This opening segment most likely starts in lower and mid-montane country where open pine and drier slopes dominate, then rises toward cooler forest around Harlan. Hikers should watch for the first clear switch from park-like spacing to denser canopy and for the way north-facing folds stay greener longer than nearby south-facing slopes. The educational point is simple and foundational: **two slopes at the same elevation can behave like different ecosystems.** 23

Harlan to Devil's Wash Basin

This is a good day to focus on mid-elevation forest structure. Expect a stronger mixed-conifer signal, especially in shaded drainages and basin margins, with more luxuriant understory where light reaches the ground. If the group sees aspen patches or richer herb cover, that is the moment to explain how cold-air drainage, snow persistence, and deeper local soils create pockets of biological intensity inside otherwise conifer-dominated terrain. 24

Devil's Wash Basin to Cimarron River

The descent toward a river or major perennial corridor should feel ecologically obvious. Expect vegetation to bunch around water: taller shrubs, broaderleaf trees where suitable alluvium exists, denser herb cover, and a noticeable cooling or humidity effect in the corridor itself. Have hikers compare the riparian strip to the surrounding slopes and ask what changed first—temperature, soil depth, or water availability. The correct answer is usually all three together. 25

Cimarron River to Santa Claus

Climbing away from the river should re-open the gradient in reverse. Watch for cottonwood/willow influence fading, then more open pine, then cooler mixed-conifer tendencies as elevation increases. This is a strong teaching day for “riparian exceptionalism”: the stream corridor is not just one more plant community; it is a hydrologic override that temporarily reshuffles the normal zonation rules. 26

Santa Claus to Baldy Town

This is where the trek enters its clearest high-country phase. Expect aspen, cooler conifer, meadow pockets, and a larger role for seeps and shallow wet ground. Around Baldy Town and adjacent high terrain, hikers should start looking for the subalpine mosaic rather than a single forest type: spruce-fir-like shade, aspen clones in brighter patches, and wet meadow vegetation in flat or poorly drained pockets. Wildlife attention shifts here too—elk sign, hummingbirds in bloom-rich openings, and high-country bird species all become more plausible. 27

Baldy layover and Baldy ascent

This is the single best natural-history day of the trek. The climb toward Baldy teaches treeline better than any lecture can. Trees become shorter, then more wind-shaped, then stop. Above that, the plant world compresses to low turf, sedges, scattered alpine flowers, lichen-spotted rock, and exposed soil. This is also the best zone to discuss why growing seasons shrink, why soils are thinner, and why many high-country

species survive by staying physically small and close to the ground. Nearby southern Rockies high-country sources strongly support this pattern and make Baldy one of Philmont's best ecological classrooms. 28

Baldy Town to Pueblano via French Henry

This is the best day to tie natural history to geology and mining disturbance. French Henry and adjacent high mining terrain combine subalpine ecology with heavily modified ground, roadcuts, tailings, and altered drainage. Hikers should compare vegetation on intact meadow soils versus disturbed mine substrate: the latter is typically patchier, rockier, and slower to carry the same plant cover. Even where the precise plant list shifts year to year, the ecological lesson is stable—substrate disturbance changes water-holding capacity, rooting opportunity, and successional pace. 29

Pueblano to Elkhorn

This is where the north-country burn story may begin to become legible in a more obvious way, depending on the exact tread and recent route conditions. Expect mosaics: some surviving forest, some open recovery patches, some shrub/aspen-dominated regeneration, and some snag-rich areas. Teach hikers to read fire severity from structure rather than from a label: scattered survivors with grassy understory usually suggest lower severity; broad snag fields and channel incision suggest hotter burn or stronger post-fire flood effects. 6

Elkhorn to Ponil

As the route drops, warmth and dryness should reassert themselves. Forest opens, grass and shrub cover become more conspicuous, and south-facing exposures begin to dominate the feel of the day. This is also a good segment for spotting how post-fire landscapes recover unevenly: places with better soil storage and cooler aspect rebound differently from shallow sunny slopes. 30

Ponil to Indian Writings

This is one of the best combined ecology-and-culture days on the trek. You are back in lower-elevation, drier country where open woodland, rocky substrate, and cultural features coexist. Indian Writings is not just an archaeological stop; it also sits in a lower, warmer ecological setting that contrasts sharply with Baldy. Students should notice how the vegetation is more open, the soils often thinner and rockier, and the whole landscape structurally simpler than the high subalpine forests earlier in the trek. 31

Field-ready species guide and practical use

A compact, field-use species list is more valuable on trek than an exhaustive checklist. The table below is intentionally selective and focuses on species or species-groups that teach habitat well. Conservation status is included only where I have reasonably high-confidence support from the sources reviewed; otherwise I mark the taxon as common or regionally characteristic rather than pretending to precision I do not have.

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Taxon	Group	Most likely zone on Trek 12-15	Season / visibility	Conservation note
Ponderosa pine	Tree	Lower to mid montane	Year-round	Common, characteristic
Douglas-fir / white fir mix	Tree	Mid-elevation cooler slopes	Year-round	Common, characteristic
Quaking aspen	Tree	Mid- to high-elevation patches	Best in leaf June–Sept; color in fall	Common, characteristic
Engelmann spruce / subalpine fir complex	Tree	High subalpine	Year-round	Characteristic of highest forests
Cottonwood / willow riparian strip	Tree-shrub	River and wet draws	Most obvious in leaf and near water	Ecologically sensitive corridor
New Mexico locust and edge shrubs	Shrub	Openings, disturbances, warm edges	Late spring–summer	Good disturbance indicator
Elk	Mammal	Meadows, forest edges, high country and north-country valleys	Dawn/dusk best	Habitat management priority regionally
Mule deer	Mammal	Broadly distributed	Dawn/dusk best	Common, characteristic
Black bear	Mammal	Broadly distributed forested terrain	Usually by sign, not sight	Food-storage safety issue
Pika / marmot	Mammal	Highest rock/talus near Baldy	Daytime in fair weather	High-elevation indicator
Mountain chickadee	Bird	Conifer forest	Common by sound	Characteristic conifer bird
Steller's jay / Clark's nutcracker	Bird	Conifer to high subalpine	Common to locally likely	Strong habitat indicators
Broad-tailed hummingbird	Bird	Flower-rich meadows and aspen/spruce edges	Summer	Excellent pollination indicator
Rio Grande cutthroat trout	Fish	Sensitive cold headwaters regionally	Stream-specific	Native conservation flagship

This table synthesizes the Carson/Valle Vidal high-country setting, adjacent wilderness species summaries, and species-habitat information for hummingbirds and high-country birds. 33

A practical month-by-month phenology guide for this route is as follows. In **June**, lower slopes are usually ahead of high country; edge shrubs and many streamside plants are active first. In **July**, mixed-conifer openings and subalpine meadows are usually at or near peak interest, and this is the best overall month for flower-and-bird interpretation. In **August**, monsoon moisture can refresh bloom, especially in meadows and burn openings, while insect activity and afternoon weather become more consequential. In **September**, flowers fade but aspens and broadleaf shrubs become the star attraction where the season extends that late. This is an inference from the route's elevation gradient and the regional southern-Rockies ecosystem structure, not a week-specific forecast. ³⁴

Safety, Leave No Trace, and educational use

The biggest natural-history safety issue on Trek 12-15 is not an animal; it is the combination of altitude, exposure, and weather. High-country environments around Baldy are windy, cold by comparison, and biologically short-seasoned for a reason. If hikers are above treeline, they are in a place shaped by frequent frost, violent storms, and shallow soils. The ecological reading and the safety reading are the same reading. When the trees stop, shelter stops too. ³⁵

Leave No Trace priorities should follow ecological sensitivity rather than generic slogans. In riparian areas, the main rule is to avoid widening social trails and trampling banks. In high meadows, spread-out lounging does more damage than most hikers realize, because the growing season is short and soils are often waterlogged or shallow. In burn-recovery areas, avoid treating young aspen or shrub thickets as off-trail shortcuts; these are exactly the places where succession is most visibly underway. Around archaeological areas such as Indian Writings, the correct stance is even stricter: no touching, no tracing, no "better photo angle" scrambling on panels or associated features. ³⁶

If you want this guide to teach, not just inform, build each day around one question. On the Harlan and Santa Claus days: **Why is this slope greener than the one across from it?** On the Baldy day: **Where exactly does treeline begin, and why here?** On the French Henry day: **What changed because of mining substrate and what changed because of elevation?** On the Ponil/Indian Writings days: **Why do ecology and human settlement/art both concentrate in certain corridors?** Those questions are stronger than species bingo because they train hikers to read landscape processes. ³⁷

Citizen-science opportunities are best kept lightweight so they do not slow the crew. The most useful ones are repeat-photo points at major viewpoints, meadow wildflower photo logs keyed to elevation, simple wildlife-sign notes rather than only direct sightings, and burn-recovery comparison photographs in north-country openings. A very good practice is to have one hiker photograph the same kind of slope each day—south-facing dry slope, north-facing forested slope, stream corridor, and high meadow—so the crew comes home with a visual gradient rather than random scenery. That produces a better custom guide later. ³⁸

Open questions and limitations

Some parts of this synthesis are stronger than others. The elevation framework, major vegetation communities, Baldy high-country ecology, elk significance, and post-fire north-country context are all well supported by the sources above. More granular items—especially exact camp-by-camp reptile, amphibian, insect, invasive-species, and flower inventories for your exact departure week—are less secure because I did not find a current, publicly indexed Philmont-specific biological inventory matched to Trek 12-15 in the

sources reviewed. Where necessary, I have labeled those items as **likely** or as landscape-based inference rather than presenting them as confirmed records. ³⁹

The cleanest way to strengthen the next version of this custom guide would be to add one or more of the following source types: a recent Philmont natural-resource or conservation report, a current GIS vegetation layer for the ranch, a recent camp-by-camp flora/fauna handout if Philmont issues one internally, and any crew photos from prior years on the same itinerary. With those, the broad framework here could be turned into a much more precise field guide.

¹ ³ ⁵ ⁷ ⁸ ¹¹ ¹² ¹³ ²⁰ ²¹ ²² ²³ ³¹ ³⁴ ³⁷ ³⁸ ³⁹ Home | Carson National Forest | Forest Service

<https://www.fs.usda.gov/r03/carson>

² ¹⁴ ¹⁹ ²⁵ ²⁶ ³² ³³ Carson National Forest | Valle Vidal | Forest Service

<https://www.fs.usda.gov/r03/carson/recreation/valle-vidal>

⁴ ³⁰ Carson National Forest | Colorado Borderlands | Forest Service

<https://www.fs.usda.gov/r03/carson/recreation/colorado-borderlands>

⁶ ³⁶ Ponil Complex Fire

https://en.wikipedia.org/wiki/Ponil_Complex_Fire?utm_source=chatgpt.com

⁹ Robinia neomexicana

https://en.wikipedia.org/wiki/Robinia_neomexicana?utm_source=chatgpt.com

¹⁰ ¹⁷ Broad-tailed hummingbird

https://en.wikipedia.org/wiki/Broad-tailed_hummingbird?utm_source=chatgpt.com

¹⁵ Carson National Forest | Valle Vidal Annual Closures | Forest Service

<https://www.fs.usda.gov/r03/carson/alerts/valle-vidal-annual-closures>

¹⁶ Mountain chickadee

https://en.wikipedia.org/wiki/Mountain_chickadee?utm_source=chatgpt.com

¹⁸ ²⁷ ²⁸ ³⁵ Wheeler Peak Wilderness

https://en.wikipedia.org/wiki/Wheeler_Peak_Wilderness?utm_source=chatgpt.com

²⁴ Columbine-Hondo Wilderness - Wikipedia

https://en.wikipedia.org/wiki/Columbine%E2%80%93Hondo_Wilderness

²⁹ Baldy Mountain (Colfax County, New Mexico)

https://en.wikipedia.org/wiki/Baldy_Mountain_%28Colfax_County%2C_New_Mexico%29?utm_source=chatgpt.com